MDPI

Article

Evolución del Estado de bienestar de los jóvenes europeos

Ignacio Lacruz Mascarós, Nicolás Camacho Albiñana

* Correspondence:

Simple Summary: A Simple summary goes here.

Abstract: A single paragraph of about 200 words maximum. For research articles, abstracts should give a pertinent overview of the work…

Cita
Evo
del
Esta
de
bien
de
los
jóve
euro
Jour
Not
Spec
202
1,
0.
https://doi.org/

0.1. Importación de los DataFrames

```
library(readr)
library(stringr)
library(dplyr)
library(tidyr)
library(magrittr)
library(lubridate)
library(ggplot2)
library(GGally)
```

```
# Cargamos el dataframe
poblacion <- read_tsv('data/poblacion_paises.tsv')

# Limpiamos los nombres de la variable country para obtener los nombres de los pai
colnames(poblacion)[1] <- 'country'
poblacion %<>% mutate(country = sub("^.*,", "", country)) %>% mutate(across(-1, ~

# Calculamos la media poblacional de cada pais desde el 2000 hasta hoy
poblacion %<>% transmute(country, population = rowMeans(select(., -country), na.rm

# Ordenamos los paises de mayor a menor poblacion
# Eliminamos algunos que no pertenecen politicamente a Europa
poblacion %<>% arrange(desc(population)) %>% filter(!(country %in% c('RU', 'UA', '

# Nos quedamos con los nombres de los 10 primeros
mas.poblacion10 <- poblacion$country[1:10]

# Nos quedamos con los nombres de los 5 primeros
mas.poblacion5 <- poblacion$country[1:5]
```

```
# Cargamos el dataframe
indep.resid <- read_tsv('data/edad_independencia_residencial.tsv')

# La primera columna viene con los valores de sex y country, por lo que hay que se
colnames(indep.resid)[1] <- 'sex.country'
indep.resid %<>% mutate(sex.country = sub("^.*,([^,]*,[^,]*)$", "\\1", sex.country
```


Received:

```
# Aqui pasamos los valores de edad de emancipacion en funcion de año a formato tid
# Filtramos para quedarnos con sex = 'T', el total, ya que no nos interesa la dife
# Nos quedamos con los 10 paises mas poblados
indep.resid %<>% mutate(across(c(-1, -2), ~ as.numeric(gsub('[^0-9.]', '', .x))))
```

```
precio.vivienda <- read_tsv('data/precio_vivienda.tsv')

colnames(precio.vivienda)[1] <- 'country'
precio.vivienda %<>% mutate(country = sub(".*,", "", country))

precio.vivienda %<>% mutate(across(-1, ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%

# Como el indice viene por cuatrimestres, nos quedamos con 1 solo valor, el corres
precio.vivienda %<>% separate(year.quadri, into = c('year', 'quadri'), sep = '-')
```

```
precio.alquiler <- read_tsv('data/precio_alquiler.tsv')

colnames(precio.alquiler)[1] <- 'country'
precio.alquiler %<>% mutate(country = sub(".*,", "", country))

precio.alquiler %<>% mutate(across(-1, ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
```

```
salarios2 <- read_tsv('data/salarios2.tsv')

# Este dataframe ha sido obtenido de otra fuente
# Por eso los nombres de los paises vienen en otro formato y hay que hacer el filt
salarios2 %<>% transmute(country = ref_area.label, year = time, earnings = 0.9*obs
names.country <- c('Germany', 'France', 'Italy', 'Spain', 'Poland', 'Romania', 'Ne
names.country %in% salarios2$country
salarios2 %<>% filter(country %in% names.country) %>% mutate(country = factor(coun

# En algunos paises se ha cambiado el metodo de cuantificar los salarios, por lo q
# Identificamos los paises en los que ha pasado esto
major.source <- salarios2 %>% count(country, source) %>% group_by(country) %>% arr

# Para no estropear la tendencia temporal, nos quedamos con los datos correspondie
salarios2 %<>% full_join(major.source) %>% mutate(earnings = ifelse(source == majo
```

0.2. Datos ausentes 5

```
# Juntamos todas las variables en un dataframe por paises y años
jovenes.df <- indep.resid %>% full_join(salarios2) %>% full_join(precio.alquiler)

## Joining with `by = join_by(country, year)`                                    6
## Joining with `by = join_by(country, year)`                                    7
## Joining with `by = join_by(country, year)`                                    8
```

```
dim(jovenes.df)

## [1] 269   6                                                                   9
```

```r
# Numero de registros y de variables

vars.num <- setdiff(colnames(jovenes.df)[sapply(jovenes.df, is.numeric)], 'year')
vars.num
```

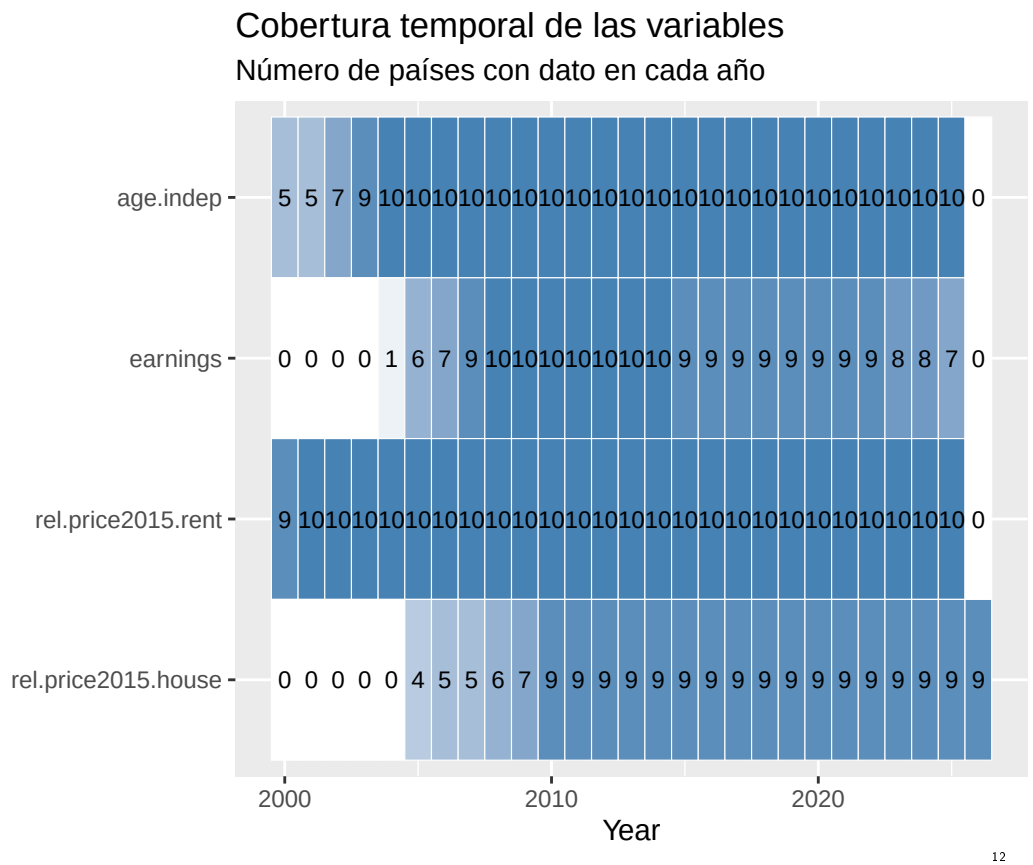
```
## [1] "age.indep"           "earnings"            "rel.price2015.rent"
## [4] "rel.price2015.house"
```

```r
# Variables cuantitativas continuas. 'country' es cualitativa nominal y 'year' es

por.var <- function(f) unname(sapply(jovenes.df[vars.num], f))
# Aplica una funcion a cada variable numerica y devuelve un vector sin nombres
```

```r
cobertura <- jovenes.df %>% group_by(year) %>% summarise(across(all_of(vars.num),

ggplot(cobertura, aes(x = year, y = variable, fill = n.paises)) + geom_tile(color
```




```r
ggsave('results/univ_cobertura.pdf', width = 14, height = 10, dpi = 300)
```

```r
# Limitamos el estudio desde el 2010 hasta el 2025
jovenes.df %<>% filter(year >= 2010 & year <= 2025)

# Obtenemos el primer año en el que comienzan las medidas de salarios y precios
# Tomamos este año como valor de referencia para poder compararlos luego mas facil
year.start <- jovenes.df %>% group_by(country) %>% arrange(year) %>% select(c(coun

jovenes.df <- full_join(jovenes.df, year.start)
```

```
## Joining with `by = join_by(country)`
```

```
# Ahora tomamos, para cada variable, un valor de 100 en el año de referencia, asi
jovenes.df %<>% group_by(country) %>% mutate(earnings = earnings/earnings[year ==

# Calculamos los salarios en proporcion al precio de la vivienda y a los alquilere
jovenes.df %<>% mutate(earn.prop.house = earnings/price.house*100, earn.prop.rent

# Calculamos el aumento porcentual de los salarios, alquileres y precio de la vivi
jovenes.df %<>%
  group_by(country) %>%
  arrange(year, .by_group = TRUE) %>%
  mutate(
    age.diff = age.indep - lag(age.indep),
    earn.diff = earnings - lag(earnings),
    house.diff = price.house - lag(price.house),
    rent.diff = price.rent - lag(price.rent)
  ) %>%
  ungroup()
```

```
# Calculamos ahora cuales son las variables cuantitativas continuas tras las trans
vars.num <- setdiff(colnames(jovenes.df)[sapply(jovenes.df, is.numeric)], 'year')
vars.num
```

```
##  [1] "age.indep"       "earnings"        "price.rent"      "price.house"
##  [5] "earn.prop.house" "earn.prop.rent"  "age.diff"        "earn.diff"
##  [9] "house.diff"      "rent.diff"
```

0.3. Estudio de variables

```
sumario.jovenes <- data.frame(variable = colnames(jovenes.df))
sumario.jovenes$type <- sapply(jovenes.df, function(x) class(x)[1])
sumario.jovenes$levels <- sapply(jovenes.df, function(x) length(unique(x)))
sumario.jovenes$topLevel <- sapply(jovenes.df, function(x) names(which.max(table(x
sumario.jovenes$topCount <- sapply(jovenes.df, function(x) max(table(x)))
sumario.jovenes$topFrac <- sumario.jovenes$topCount / nrow(jovenes.df)
sumario.jovenes$missFreq <- sapply(jovenes.df, function(x) sum(is.na(x)))
sumario.jovenes$missFrac <- sumario.jovenes$missFreq / nrow(jovenes.df)

sumario.jovenes
```

```
##           variable    type levels          topLevel topCount topFrac missFreq
## 1          country  factor     10                DE       16 0.10000        0
## 2             year numeric     16              2010       10 0.06250        0
## 3        age.indep numeric     61              23.7       12 0.07500        0
## 4         earnings numeric    128               100        9 0.05625        0
## 5       price.rent numeric    136               100        9 0.05625        0
## 6      price.house numeric    136               100        9 0.05625        0
## 7  earn.prop.house numeric    128               100        9 0.05625        0
## 8   earn.prop.rent numeric    128               100        9 0.05625        0
## 9         age.diff numeric     29                 0       31 0.19375       10
## 10       earn.diff numeric    127 -26.0178668438882        1 0.00625        4
## 11      house.diff numeric    134  1.45547945205479        3 0.01875        5
## 12       rent.diff numeric    132 0.210084033613441        2 0.01250        5
```

```
##    missFrac
## 1   0.00000
## 2   0.00000
## 3   0.00000
## 4   0.15625
## 5   0.10000
## 6   0.10000
## 7   0.15625
## 8   0.15625
## 9   0.06250
## 10  0.21250
## 11  0.15625
## 12  0.15625
```

```
cobertura.var <- data.frame(variable = vars.num)
cobertura.var$year.min <- unname(sapply(vars.num, function(v) min(jovenes.df$year[
cobertura.var$year.max <- unname(sapply(vars.num, function(v) max(jovenes.df$year[
cobertura.var$n.obs <- unname(sapply(vars.num, function(v) sum(!is.na(jovenes.df[[
cobertura.var$n.paises <- unname(sapply(vars.num, function(v) length(unique(jovene
cobertura.var
```

```
##           variable year.min year.max n.obs n.paises
## 1        age.indep     2010     2025   160       10
## 2         earnings     2010     2025   135        9
## 3       price.rent     2010     2025   144        9
## 4      price.house     2010     2025   144        9
## 5  earn.prop.house     2010     2025   135        9
## 6   earn.prop.rent     2010     2025   135        9
## 7         age.diff     2011     2025   150       10
## 8        earn.diff     2011     2025   126        9
## 9       house.diff     2011     2025   135        9
## 10       rent.diff     2011     2025   135        9
```

0.4. Análisis univariante

0.4.1. Estadísticos descriptivos de las variables numéricas

```
mystats <- function(x, na.omit = TRUE) {
  if (na.omit) x <- x[!is.na(x)]
  m <- mean(x)
  n <- length(x)
  s <- sd(x)
  skew <- sum((x - m)^3 / s^3) / n
  kurt <- sum((x - m)^4 / s^4) / n - 3
  return(c(n = n, mean = m, trimmed = mean(x, trim = 0.1), median = median(x), min
}
# Funcion del Tema 6 (asimetria y curtosis en exceso), ampliada con media recortad

estad <- sapply(jovenes.df[vars.num], mystats)

t(round(estad[c('n', 'mean', 'trimmed', 'median', 'min', 'q1', 'q3', 'max'), ], 2)
```

```
##                   n   mean trimmed median    min     q1     q3    max
## age.indep       160  26.77   26.74  27.00  23.00  23.90  29.00  30.90
## earnings        135 121.47  109.75 103.22  64.26  95.58 118.49 431.60
```

```
## price.rent      144 115.02  111.73 108.71  99.46 104.36 118.98 190.88
## price.house     144 116.87  111.63 106.05  71.00  94.69 130.36 247.48
## earn.prop.house 135 106.72   98.67 100.00  63.85  83.69 112.76 330.68
## earn.prop.rent  135 103.76   96.97  97.48  64.35  86.01 105.86 258.22
## age.diff        150   0.02    0.02   0.00  -1.80  -0.10   0.20   1.80
## earn.diff       126   6.04    3.72   4.08 -26.02  -3.03  10.20  64.47
## house.diff      135   4.20    3.71   3.47 -13.68   0.46   6.87  43.17
## rent.diff       135   2.86    2.23   1.81  -1.30   1.15   3.17  19.45
```

```
# Tendencia central y posicion

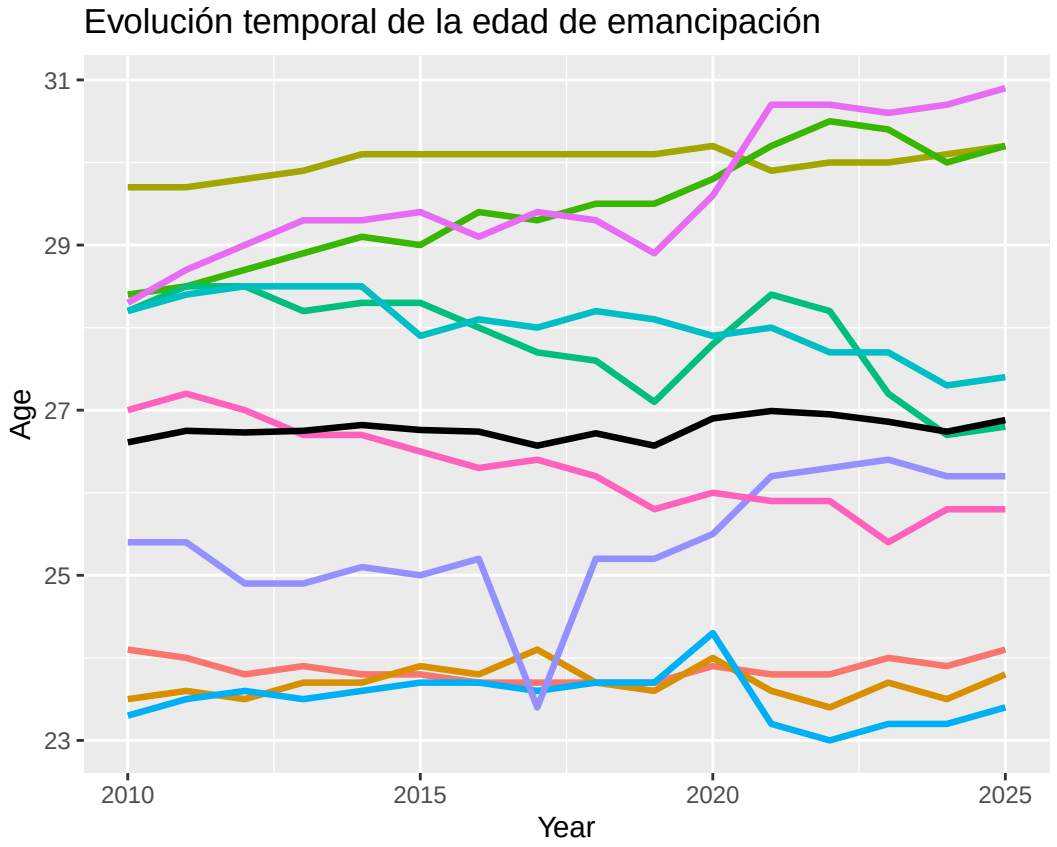
t(round(estad[c('stdev', 'var', 'mad', 'IQR', 'skew', 'kurtosis'), ], 2))
```

```
##                 stdev     var   mad   IQR  skew kurtosis
## age.indep        2.49    6.21  3.56  5.10 -0.03    -1.48
## earnings        55.57 3087.57 15.46 22.91  3.03    10.57
## price.rent      17.18  295.04  8.92 14.62  2.06     4.55
## price.house     34.21 1170.42 22.36 35.67  1.48     2.01
## earn.prop.house 42.96 1845.74 22.56 29.07  3.11    10.71
## earn.prop.rent  33.83 1144.40 15.12 19.85  2.74     8.00
## age.diff         0.35    0.12  0.30  0.30 -0.18     8.67
## earn.diff       16.95  287.13  9.41 13.24  1.68     3.61
## house.diff       7.66   58.70  4.87  6.41  1.35     5.13
## rent.diff        3.31   10.96  1.31  2.02  2.63     8.27
```

```
# Dispersion y forma (skew > 0: cola larga a la derecha; kurtosis > 0: colas mas p
```

0.4.2. Edad de emancipación

```
age.mean <- jovenes.df %>% group_by(year) %>% summarise(age.mean = mean(age.indep,
ggplot() + geom_line(data = jovenes.df, mapping = aes(x = year, y = age.indep, col
  scale_color_manual(values = c(setNames(scales::hue_pal()(length(unique(jovenes.d
  labs(title = 'Evolución temporal de la edad de emancipación') + xlab('Year') + y
```




```
ggsave('results/edad_emancipacion.pdf', width = 14, height = 10, dpi = 300)
```

```
media.edad <- jovenes.df %>% group_by(country) %>% summarise(n.years = n(), media
media.edad
```

```
##    country n.years media media.pct total.dif
## 1       DE      16 23.86      0.00       0.0
## 2       FR      16 23.69      0.02       0.3
## 3       IT      16 30.01      0.03       0.5
## 4       ES      16 29.46      0.12       1.8
## 5       PL      16 27.84     -0.09      -1.4
## 6       RO      16 28.02     -0.05      -0.8
## 7       NL      16 23.51      0.01       0.1
## 8       BE      16 25.41      0.05       0.8
## 9       EL      16 29.62      0.17       2.6
## 10      CZ      16 26.29     -0.08      -1.2
```

```
# Media anual de cada pais de la edad de emancipación
```

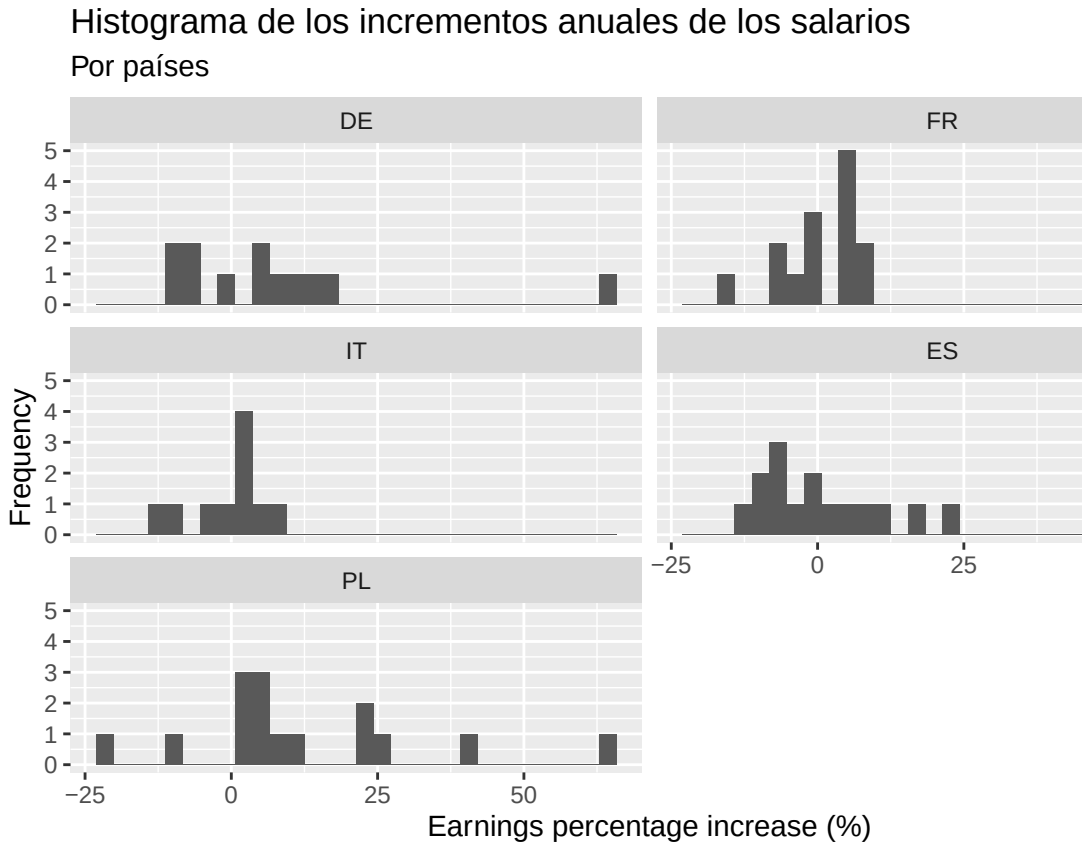
0.4.3. Media del aumento anual por país: salarios, alquiler y vivienda

```
## year.ini year.fin
##     2010     2020
```

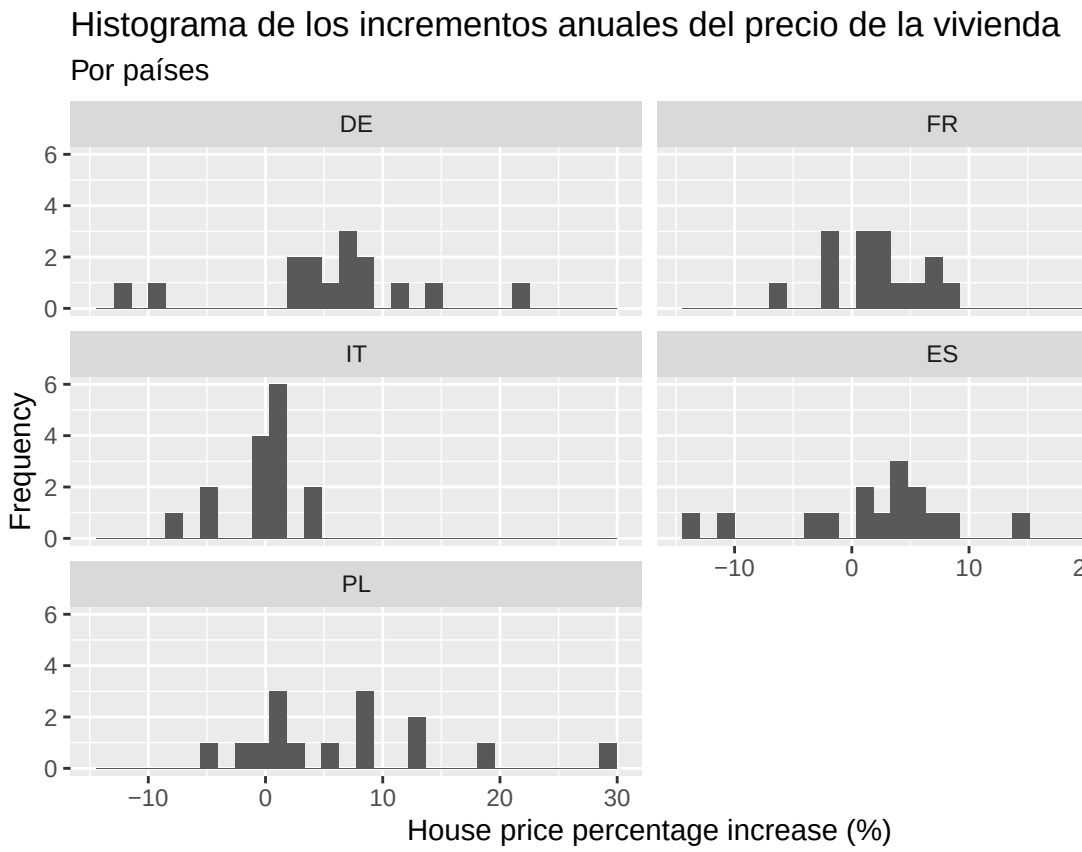
```
##   country  n
## 1      DE 10
## 2      FR 10
## 3      IT 10
```

```
## 4       ES 10
## 5       PL 10
## 6       RO 10
## 7       NL 10
## 8       BE 10
## 9       CZ 10
```

```
jovenes.df %>% filter(country %in% mas.poblacion5) %>% ggplot(aes(x=earn.diff)) +
```

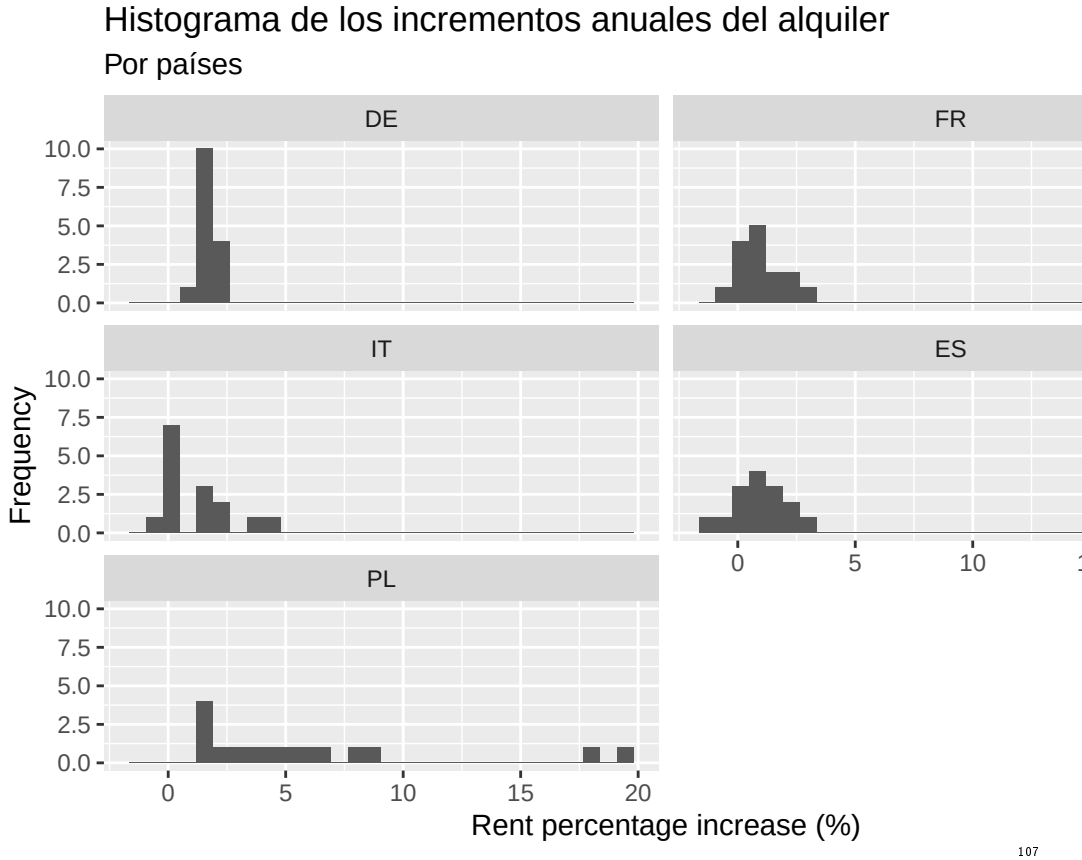



```
jovenes.df %>% filter(country %in% mas.poblacion5) %>% ggplot(aes(x=house.diff)) +
```



106

```
jovenes.df %>% filter(country %in% mas.poblacion5) %>% ggplot(aes(x=rent.diff)) +
```



107

```r
medias.pais <- aum.comun %>% group_by(country) %>% summarise(n.years = n(), across
medias.pais %>% mutate(across(-c(country, n.years), ~ round(.x, 2)))
```

```
##   country n.years house.pct rent.pct earn.pct earn.menos.house earn.menos.rent
## 1      DE      10      6.12     1.47     8.28             2.16            6.81
## 2      FR      10      1.78     0.73    -0.09            -1.87           -0.83
## 3      IT      10     -1.49     0.65    -0.34             1.15           -0.99
## 4      ES      10     -0.53     0.39    -1.13            -0.60           -1.52
## 5      PL      10      2.59     3.15     6.69             4.10            3.53
## 6      RO      10      0.46     1.76    13.49            13.04           11.73
## 7      NL      10      2.47     2.99     0.13            -2.35           -2.87
## 8      BE      10      2.79     1.44     0.31            -2.48           -1.13
## 9      CZ      10      5.65     2.66     4.36            -1.29            1.70
```

```r
# Media anual de cada pais en la ventana comun y diferencias entre variables

pares <- list(c('earn.pct', 'house.pct'), c('earn.pct', 'rent.pct'))

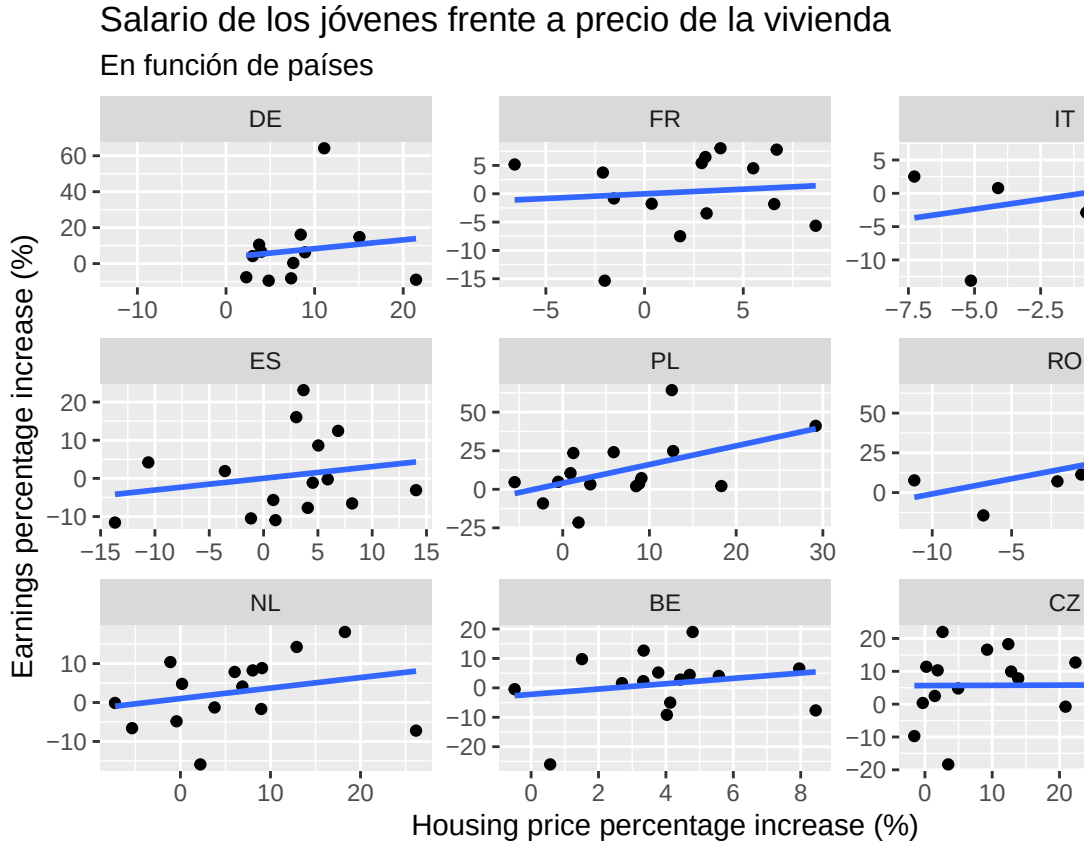
data.frame(par = sapply(pares, paste, collapse = ' vs '), paises.mayor = sapply(pa
```

```
##                     par paises.mayor n.paises
## 1 earn.pct vs house.pct            4        9
## 2  earn.pct vs rent.pct            4        9
```

```r
# paises.mayor: numero de paises en los que la media de la primera variable es may
```

0.5. Análisis bivariante

```r
jovenes.df %>% filter(country != 'EL') %>% ggplot(aes(x = house.diff, y = earn.dif
   geom_smooth(method = "lm", se = FALSE, na.rm = TRUE) + facet_wrap(.~country, nr
```




```
jovenes.df %>% filter(country != 'EL') %>% group_by(country) %>% summarise(cor.pea
```

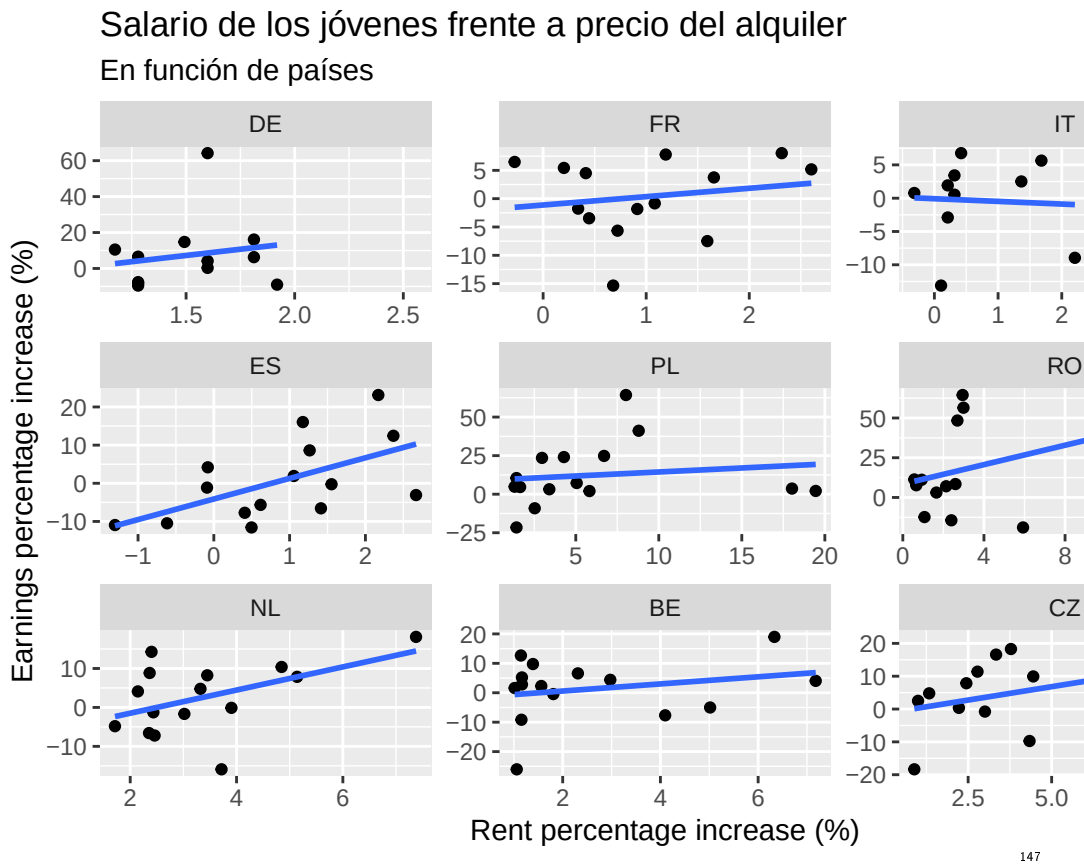
```
## # A tibble: 9 x 3
##   country cor.pearson cor.spearman
##   <fct>         <dbl>        <dbl>
## 1 DE            0.136        0.217
## 2 FR            0.101        0.134
## 3 IT            0.272        0.345
## 4 ES            0.206        0.232
## 5 PL            0.517        0.386
## 6 RO            0.344        0.175
## 7 NL            0.260        0.3
## 8 BE            0.205        0.207
## 9 CZ            0.00748      0.104
```

```
jovenes.df %>% filter(country != 'EL') %>% group_by(country) %>% summarise(slope =
```

```
## # A tibble: 9 x 2
##   country   slope
##   <fct>     <dbl>
## 1 DE      0.489
## 2 FR      0.164
## 3 IT      0.579
## 4 ES      0.307
## 5 PL      1.21
## 6 RO      1.91
## 7 NL      0.270
```

```
## 8 BE      0.905
## 9 CZ      0.00674
```

```
jovenes.df %>% filter(country != 'EL') %>% ggplot(aes(x = rent.diff, y = earn.diff
  geom_smooth(method = "lm", se = FALSE, na.rm = TRUE) + facet_wrap(.~country, nro
```




```
jovenes.df %>% filter(country != 'EL') %>% group_by(country) %>% summarise(cor.pea
```

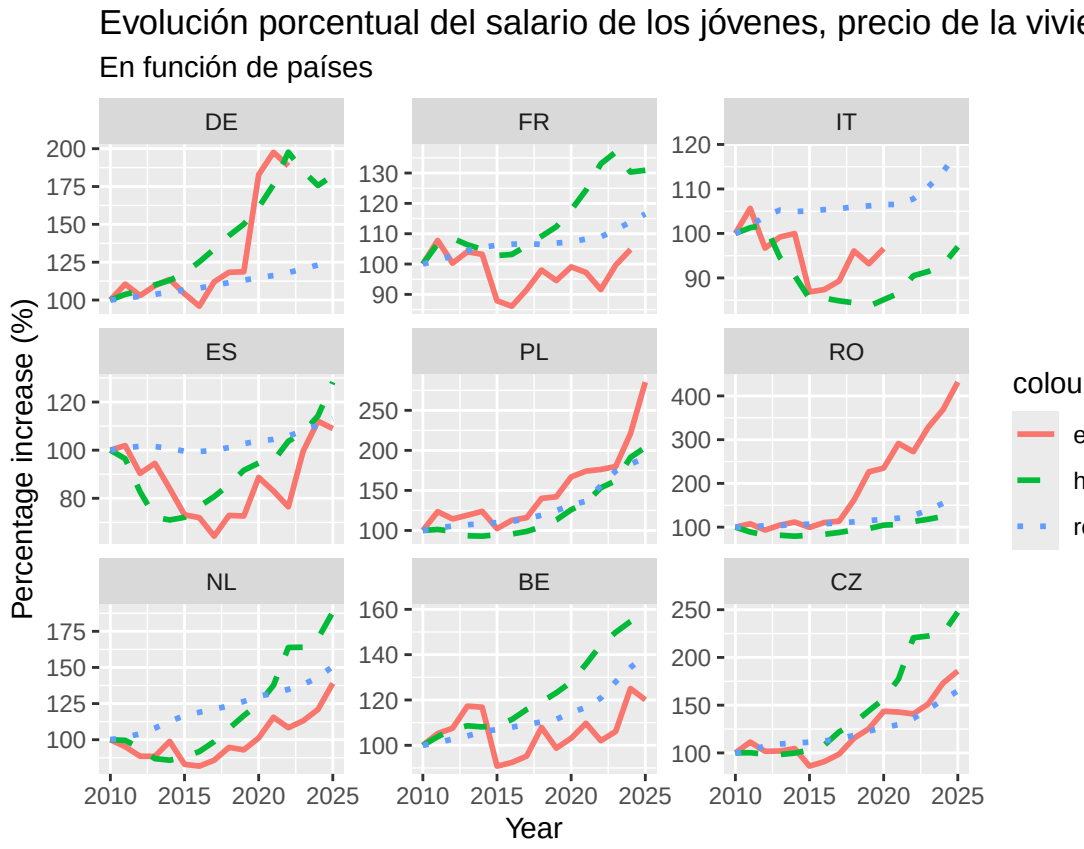
```
## # A tibble: 9 x 3
##   country cor.pearson cor.spearman
##   <fct>         <dbl>        <dbl>
## 1 DE            0.174        0.154
## 2 FR            0.178        0.0901
## 3 IT           -0.0514       0.354
## 4 ES            0.572        0.554
## 5 PL            0.142        0.236
## 6 RO            0.506        0.439
## 7 NL            0.484        0.346
## 8 BE            0.237        0.211
## 9 CZ            0.470        0.479
```

```
jovenes.df %>% filter(country != 'EL') %>% group_by(country) %>% summarise(slope =
```

```
## # A tibble: 9 x 2
##   country  slope
##   <fct>    <dbl>
## 1 DE      13.8
```

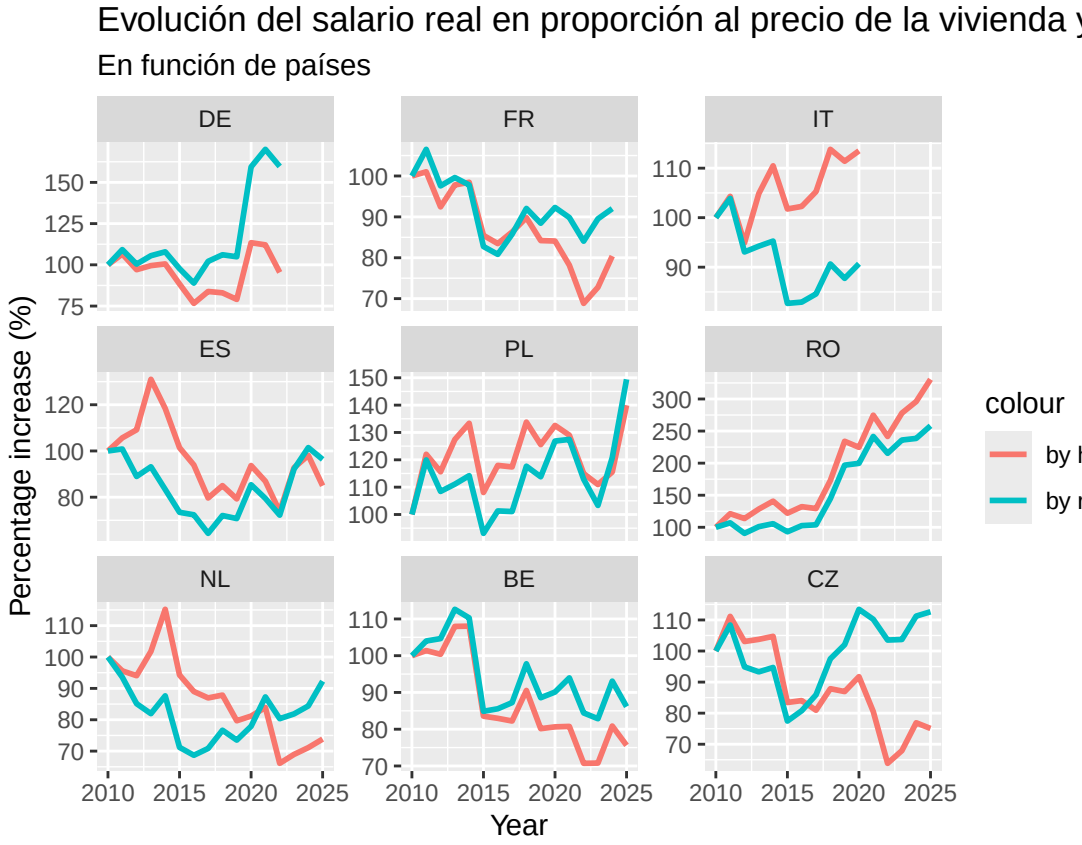
```
## 2 FR       1.48
## 3 IT      -0.401
## 4 ES       5.40
## 5 PL       0.519
## 6 RO       3.05
## 7 NL       2.97
## 8 BE       1.21
## 9 CZ       1.63
```

```
ggplot(data = filter(jovenes.df, country != 'EL', year >= 2005 & year <= 2025)) +
  geom_line(aes(x = year, y = earnings, color = 'earnings'), linewidth = 1, na.rm
  geom_line(aes(x = year, y = price.house, color = 'house buying'), linewidth = 1,
  geom_line(aes(x = year, y = price.rent, color = 'rent'), linewidth = 1, linetype
```




```
ggsave('results/viviendas_salarios.pdf', width = 14, height = 10, dpi = 300)
```

```
ggplot(data = filter(jovenes.df, country != 'EL', year >= 2005 & year <= 2025)) +
  geom_line(aes(x = year, y = earn.prop.house, color = 'by house prices'), linewid
  geom_line(aes(x = year, y = earn.prop.rent, color = 'by rent'), linewidth = 1, n
```

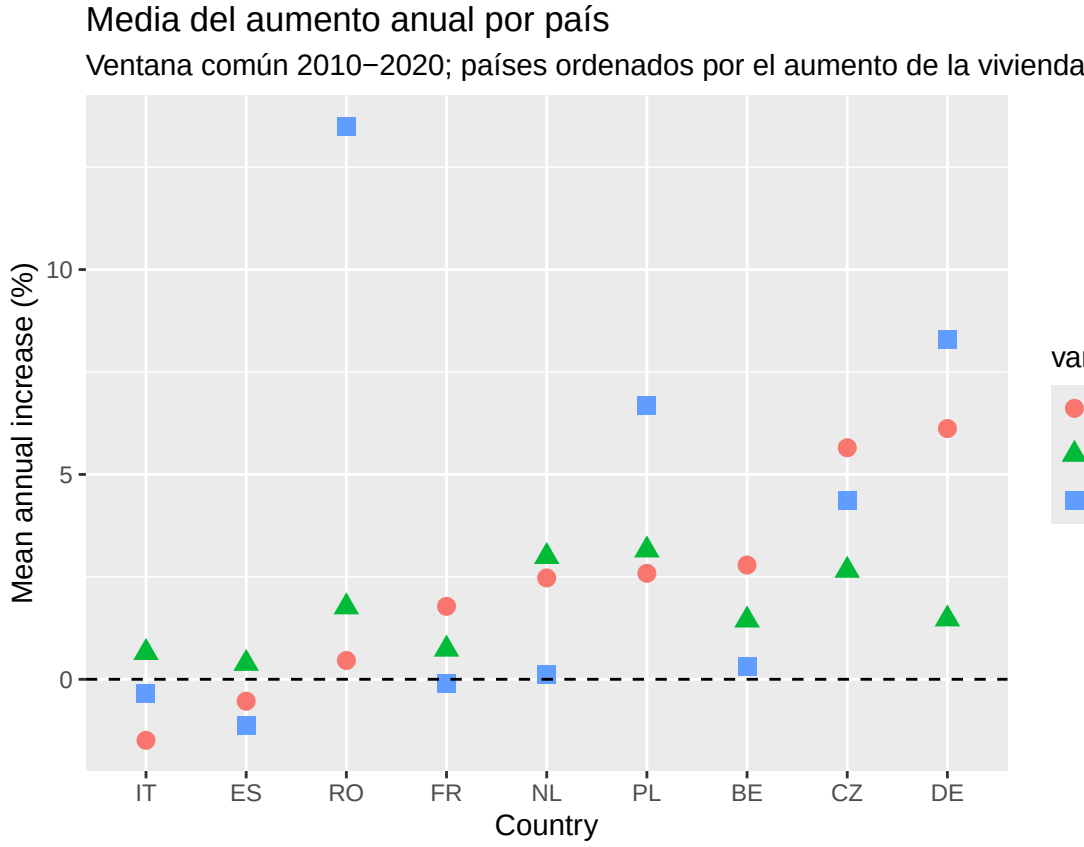


173

```
ggsave('results/proporcion_salario.pdf', width = 14, height = 10, dpi = 300)
```

```
medias.long <- medias.pais %>% pivot_longer(all_of(vars.pct), names_to = 'variable
```

```
ggplot(medias.long, aes(x = country, y = media, color = variable, shape = variable
```




```
ggsave('results/univ_aumentos_medias.pdf', width = 14, height = 10, dpi = 300)
```

Appendix A

Appendix A.1

The appendix is an optional section that can contain details and data supplemental to the main text. For example, explanations of experimental details that would disrupt the flow of the main text, but nonetheless remain crucial to understanding and reproducing the research shown; figures of replicates for experiments of which representative data is shown in the main text can be added here if brief, or as Supplementary data. Mathematical proofs of results not central to the paper can be added as an appendix.

Appendix B

All appendix sections must be cited in the main text. In the appendixes, Figures, Tables, etc. should be labeled starting with 'A', e.g., Figure A1, Figure A2, etc.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.